

TAMANRASSET, Algeria

WMO: 606800

Lat: 22.800N

Lon: 5.433E

Elev: 1364

StdP: 85.98

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.8	4.2	-20.4	0.7	9.4	-18.1	0.9	11.2	13.2	13.4	11.3	14.7	1.4	20	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	37.1	16.1	36.2	15.9	35.9	15.8	19.1	28.3	18.3	28.9	17.7	29.5	5.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.1	13.6	23.5	14.9	12.5	24.0	13.1	11.1	24.5	60.5	28.1	57.7	28.9	55.4	29.1	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.4	10.1	9.0	-0.4	39.1	1.6	1.7	-1.5	40.3	-2.4	41.3	-3.3	42.2	-4.4	43.4		
			-4.9	20.8	0.9	1.2	-5.6	21.7	-6.1	22.4	-6.6	23.1	-7.3	24.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.6	13.0	15.1	19.0	23.6	27.5	29.6	29.9	29.2	28.0	23.7	18.3	14.6
	DBStd	6.49	2.58	2.94	3.05	2.69	2.17	1.67	1.47	1.52	1.56	2.19	2.13	2.52
	HDD10.0	7	4	1	1	0	0	0	0	0	0	0	0	1
	HDD18.3	437	167	97	29	1	0	0	0	0	0	0	27	117
	CDD10.0	4623	97	144	278	407	541	588	617	596	539	424	248	144
	CDD18.3	2013	1	6	49	158	283	338	358	337	289	166	25	2
	CDH23.3	27074	27	106	626	1987	3914	4905	5204	4516	3651	1767	325	46
	CDH26.7	14203	2	19	140	855	2154	2878	3051	2496	1910	667	31	1
(14) Wind	WSAvg	3.8	3.2	3.6	4.1	4.3	4.4	4.3	4.4	4.2	3.9	3.6	3.0	3.0
(15) Precipitation	PrecAvg	42	1	1	1	1	3	6	4	11	7	3	1	1
	PrecMax	95	36	17	34	21	43	68	27	67	31	47	32	8
	PrecMin	1	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	26	5	4	5	4	8	11	7	15	8	8	5	2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.1	29.2	31.0	35.1	37.1	38.2	38.2	37.2	36.1	33.2	29.0	26.2
		MCWB	10.9	12.2	13.0	15.1	16.1	16.2	16.4	16.6	16.2	14.7	12.3	10.3
	2%	DB	23.8	26.1	29.4	33.8	36.0	37.2	37.1	36.2	35.1	32.2	27.4	24.7
		MCWB	9.2	10.8	12.0	14.3	15.7	15.9	16.1	16.2	15.8	14.5	12.0	9.9
	5%	DB	22.0	24.1	28.2	32.3	35.0	36.2	36.2	35.3	34.2	31.2	26.2	23.1
		MCWB	8.2	9.6	11.5	13.7	15.2	15.7	15.9	16.1	15.6	14.2	11.4	9.2
	10%	DB	20.5	22.9	27.0	31.1	34.1	35.5	35.5	34.8	33.8	30.1	25.2	21.9
		MCWB	7.5	8.9	11.0	13.1	14.7	15.5	15.7	16.0	15.5	13.9	10.9	8.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	12.9	14.5	16.0	18.2	19.0	19.7	20.3	19.1	17.4	14.6	12.4
		MCDB	21.3	25.1	22.7	31.0	29.9	29.6	29.3	28.3	27.3	25.2	25.5	20.5
	2%	WB	10.3	11.5	13.2	14.9	17.1	17.7	18.5	19.4	18.0	16.4	13.0	11.1
		MCDB	20.7	22.9	24.2	30.5	31.2	30.9	29.4	28.1	28.7	26.4	25.2	20.8
	5%	WB	9.0	10.4	12.3	14.1	16.2	17.0	17.7	18.7	17.3	15.7	12.2	10.1
		MCDB	20.1	22.0	25.1	30.0	31.7	31.4	30.2	28.3	29.6	27.2	24.8	21.2
	10%	WB	7.8	9.3	11.5	13.4	15.5	16.4	17.0	17.9	16.7	14.9	11.4	9.2
		MCDB	19.6	21.3	25.4	29.2	31.8	31.7	31.0	28.8	29.9	27.6	24.1	20.9
(35) Mean Daily Temperature Range	5% DB	MDBR	14.8	14.7	14.2	14.0	12.9	11.8	11.6	11.4	11.7	12.4	13.8	14.1
		MCDBR	16.3	15.7	15.0	14.6	13.2	12.7	12.4	11.9	12.2	13.1	14.5	15.1
		MCWBR	8.4	7.8	6.9	6.6	5.4	5.0	4.9	4.3	4.4	5.3	6.9	7.4
	5% WB	MCDBR	13.9	13.8	12.7	12.7	12.0	11.4	11.3	11.6	11.4	11.3	13.3	13.4
		MCWBR	7.9	7.4	6.7	6.2	5.4	4.6	4.5	4.2	4.3	4.8	6.7	7.0
(40) Clear-Sky Solar Irradiance	taub		0.288	0.317	0.398	0.510	0.592	0.641	0.551	0.579	0.542	0.472	0.369	0.313
	taud		2.442	2.347	2.052	1.774	1.619	1.547	1.723	1.689	1.756	1.899	2.167	2.366
	Ebn at Noon		986	973	901	807	738	697	763	743	762	798	878	940
	Edh at Noon		102	120	169	227	264	282	236	244	224	186	134	106
(44) All-Sky Solar Radiation	RadAvg		5.00	5.72	6.76	7.33	7.40	7.34	7.57	6.99	6.54	5.83	5.10	4.66
	RadStd		0.22	0.35	0.40	0.30	0.31	0.38	0.26	0.26	0.32	0.29	0.18	0.19

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.58	+0.97	+1.82	+3	N/A	N/A	+72
(47) Regional (2 neighbors)	N/A	N/A	N/A	+0.58	+0.97	+1.82	+3	N/A	N/A	+72

Nomenclature: See separate page